

Question 3

Not yet answered

Marked out of 2.00

Flag question

The variance of any random variable cannot be negative

Select one:

☒ True

☐ False

True

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For random variables X and Y , we have

$$\text{Cov}(\text{Var}(X), Y) =$$

Select one:

☐ a. $\text{Var}(Y)$

☐ b. 1

☐ c. 0

☐ d. $\text{Var}(X)$

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Question 5

Not yet answered

Marked out of 2.00

Flag question

For two random variables, the correlation coefficient and covariance have always the same sign.

Select one:

- ☒ True
- ☐ False

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Two orthogonal jointly wide-sense stationary processes $X(t)$ and $Y(t)$ with $\bar{x} = \bar{y} = 1$. We have $C_{XY}(t, t + \tau) = -1$

Select one:

☐ True

☐ False

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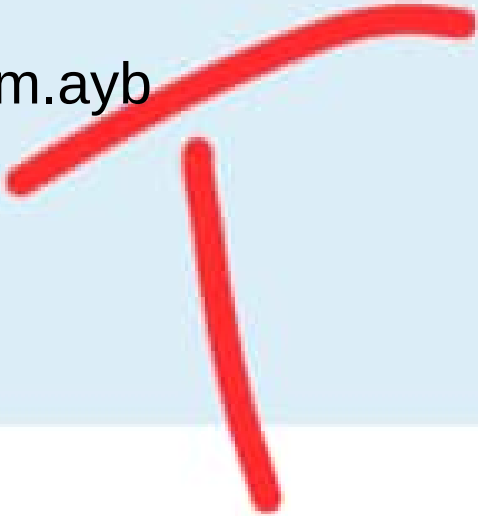
TRGP

The variance of any random variable cannot be negative

Select one:

- ☐ True
- ☒ False

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For random variables X and Y , we have

$$\text{Cov}(\text{Var}(X), Y) =$$

Select one:

☐ a. $\text{Var}(Y)$

☒ b. 0

☐ c. $\text{Var}(X)$

☐ d. 1

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Question 8

Not yet answered

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For an event A , we have $P_r(A | A) + P_r(\emptyset | A) + P_r(\Omega) =$

Select one:

- ☐ a. 4
- ☐ b. 3
- ☐ c. 1
- ☒ d. 2

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2/2

Let $X \sim \text{Poisson}(m)$. Which of the following is False about Poisson distribution

Select one:

- ☐ a. X is the number of occurrences in the interval of time
- ☐ b. the mean and variance are always equal
- ☐ c. Random variable X takes values $X = 0, 1, 2, 3, \dots$
- ☐ d. The parameter m has to be an integer number

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For an event A , we have $P_r(A | A) + P_r(\emptyset | A) + 2P_r(\Omega) =$

Select one:

- ☐ a. 4
- ☐ b. 3
- ☐ c. 2
- ☐ d. 1

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3/6

For an event A , we have $P_r(A|A) + P_r(\emptyset|A) + 2P_r(\Omega) =$

Select one:

- ☐ a. 4
- ☐ b. 3
- ☐ c. 2
- ☐ d. 1

Let X be a continuous random variable with PDF f_X . If $z = h(y)$ for some continuous function Y . Then the PDF of the transformation Y is

Select one:

- ☐ a. $f_Y(y) = (f_X \circ h)(y) |h'(y)|$
- ☐ b. $f_Y(y) = (h^{-1} \circ f_X)(y) |h'(y)|$
- ☐ c. $f_Y(y) = (h \circ f_X)(y) |h'(y)|$
- ☐ d. $f_Y(y) = (f_X \circ h^{-1})(y) |h'(y)|$

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3.3-30107353-PROBABILITY AND RANDOM VARIABLES-2 > 19 September - 25 September > Final Exam

Question 9

Not yet answered

Marked out of 1.00

Flag question

A random process $X(t)$ with power density spectrum $p_{XX}(w) = \delta(w+2) + \delta(w-2)$. The average power P_{XX} equals $\frac{1}{3\pi}$.

Select one:

- ☐ True
- ☐ False

Let X be a continuous random variable with PDF f_X . If $x = h(y)$ for some continuous function transformation Y is

Select one:

- ☐ a. $f_Y(y) = (f_X \circ h)(y) |h'(y)|$
- ☐ b. $f_Y(y) = (h^{-1} \circ f_X)(y) |h'(y)|$
- ☐ c. $f_Y(y) = (h \circ f_X)(y) |h'(y)|$
- ☐ d. $f_Y(y) = (f_X \circ h^{-1})(y) |h'(y)|$

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Question 9

Not yet answered

Marked out of 1.00

Flag question

A random process $X(t)$ with power density spectrum $p_{XX}(\omega) = \delta(\omega + 2) + \delta(\omega - 2)$. The average power P_{XX} equals $\frac{1}{2\pi}$.

Select one:

- ☐ True
- ☐ False

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Question 14

Not yet answered

Marked out of 2.00

Flag question

Two orthogonal random variables X and Y if $\bar{x} = 0.5$ and $\bar{y} = 4$, then $\text{Cov}(X, Y) =$

Select one:

☐ a. -1

☐ b. -2

☐ c. 1

☐ d. 2

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12-13 / 26

For a random variables $X \sim N(a, a)$ where a is a positive constant . If $P_r(X > 16) = 0.5$ then the standard deviation of X is

Select one:

- ☐ a. 2
- ☐ b. 4
- ☐ c. 8
- ☐ d. 16

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12

Answered

1 of 2 to

Question

A random process $X(t)$ with auto-correlation function $R_{XX}(\tau) = e^{-\tau^2}$. The process is stationary ergodic with no periodic components. Then $\bar{X} =$

Select one:

- ☐ a. 0
- ☐ b. 3
- ☐ c. 2
- ☐ d. 1

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A

For random variable $X \sim U(0, b)$ where $b > 0$ is a constant. If $E(X) = 2$ then $b =$

Select one:

- ☐ a. 4
- ☐ b. 12
- ☐ c. 8
- ☐ d. 10

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For the random process $X(t) = \cos(\omega t + \omega)$ where ω is a random variable uniformly distributed on the interval $(0, \pi)$. Then we have

Select one:

☐ a.

$$E(X(t)) = \frac{\sin(\pi(t+1))}{\pi(t+1)}$$

☐ b.

$$E(X(t)) = \frac{\cos(\pi(t+1))}{\pi}$$

☐ c.

$$E(X(t)) = \frac{\cos(\pi(t+1))}{\pi(t+1)}$$

☐ d.

$$E(X(t)) = \frac{\sin(\pi(t+1))}{\pi}$$

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For a random process with Auto-Covariance $C_{XX}(t, t + \tau) = 2e^{-2|\tau|}$, then the variance of the process equals 4

Select one:

- ☐ True
- ☐ False

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F

For an event A , we have $P_r(A | A) + P_r(\emptyset | A) + 2P_r(\Omega) =$

Select one:

☐ a. 3

☐ b. 4

☐ c. 1

☐ d. 2

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A

the alternate hypothesis H_A is what we are trying to prove

Select one:

- ☐ True
- ☐ False

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Let T be the CDF of the standard normal distribution. Then $T(1.21) = 0.90490$

Select one:

☐ True

☐ False

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F

22

answered

out of 2.00

question

$$\int_{-\infty}^{\infty} \sin(x^2) \delta(x - \sqrt{\pi}) dx =$$

Select one:

☐ a. 1

☐ b. 0.5

☐ c. -0.5

☐ d. 0

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For the random process $X(t) = A^{\omega} \cos(\omega t)$ we have ω is a positive constant and A is a random variable uniformly distributed on $(0, 1)$.
Then $E\{X(t)\} =$

Select one:

- ☐ a. $\frac{2}{\omega+1} \cos(\omega t)$
- ☒ b. $\frac{1}{\omega+1} \cos(\omega t)$
- ☐ c. $\frac{2}{\omega-1} \cos(\omega t)$
- ☐ d. $\frac{1}{\omega-1} \cos(\omega t)$

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b

Let $X \sim B(16, p)$ with $Var(X) = 3$ and $p < q$. Then $q =$

Select one:

- ☐ a. 0.5
- ☐ b. 0.25
- ☐ c. 0.2
- ☐ d. 0.75

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Let $X \sim B(16, p)$ with $Var(X) = 3$ and $p < q$. Then $p =$

Select one:

- ☐ a. 0.25
- ☐ b. 0.75
- ☐ c. 0.5
- ☐ d. 0.2

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